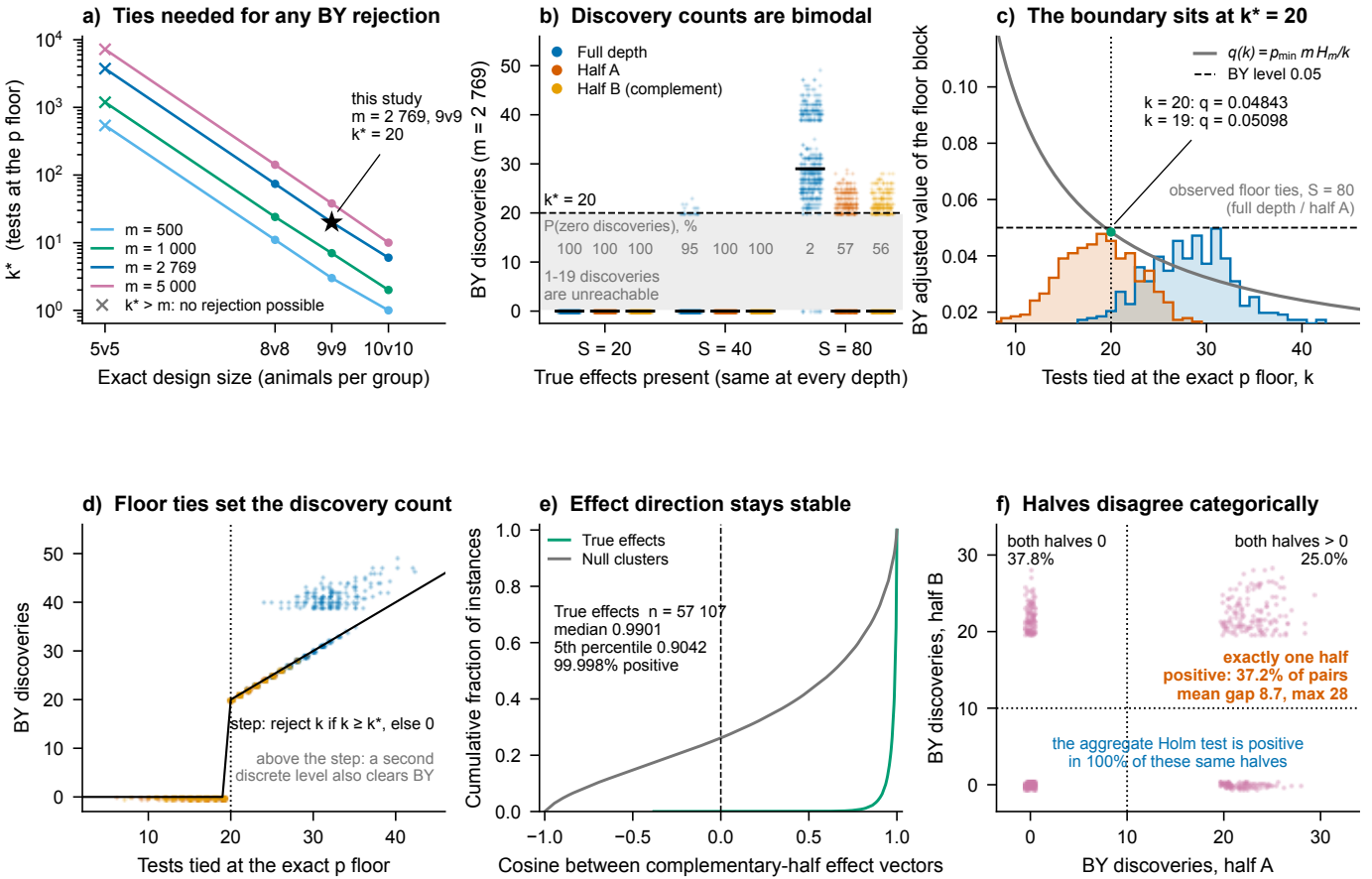

a) Ties needed for any BY rejection
k* (tests at the p floor)
this study
m = 2 769, 9v9
k* = 20
m = 500
m = 1 000
m = 2 769
m = 5 000
k* > m: no rejection possible
5v5
8v8
9v9
10v10
Exact design size (animals per group)
b) Discovery counts are bimodal
BY discoveries (m = 2 769)
Full depth
Half A
Half B (complement)
k* = 20
P(zero discoveries), %
100 100 100 95 100 100 2 57 56
1-19 discoveries
are unreachable
S = 20
S = 40
S = 80
True effects present (same at every depth)
c) The boundary sits at k* = 20
BY adjusted value of the floor block
q(k) = p_min m H_m/k
BY level 0.05
k = 20: q = 0.04843
k = 19: q = 0.05098
observed floor ties, S = 80
(full depth / half A)
Tests tied at the exact p floor, k
d) Floor ties set the discovery count
BY discoveries
step: reject k if k ≥ k*, else 0
above the step: a second
discrete level also clears BY
Tests tied at the exact p floor
e) Effect direction stays stable
Cumulative fraction of instances
True effects
Null clusters
True effects n = 57 107
median 0.9901
5th percentile 0.9042
99.998% positive
Cosine between complementary-half effect vectors
f) Halves disagree categorically
BY discoveries, half B
both halves 0
37.8%
both halves > 0
25.0%
exactly one half
positive: 37.2% of pairs
mean gap 8.7, max 28
the aggregate Holm test is positive
in 100% of these same halves
BY discoveries, half A